GlucoTwin

A glucose digital twin for Type 2 Diabetes

A model of one patient, built from their health record, CGM, wearable and meal streams. It forecasts the next two hours, gives the chance of a dangerous high or low, answers “what if they ate less”, and **states how certain it is**.

Team Synapse Squad

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Live Dashboard

https://cleopatrakilledceaser.github.io/SynapseSquad_IITGuwahati/

Happiest Health Digital Twin Challenge 2026 · opens in a browser, no install, no sign-in

The problem: the data exists, but no one can read it

Type 2 diabetes is managed in appointments weeks or months apart. Between them the patient generates more data than anyone has time to look at — and the one number the clinician does see is a **three-month average**.

101M

adults in India live with diabetes — **11.4 %** of the population.

136M

more are prediabetic — **15.3 %** — the population this tooling would reach next.

> Specialist diabetes care is concentrated in large cities, and almost no tooling is built around Indian diets.

Source: ICMR-INDIAB, *Lancet Diabetes & Endocrinology*, 2023

Two minutes, one number

1
HBA1C

What the clinician has in the room: **one HbA1c**, a handful of labs, and whatever the patient remembers about the last three months — in roughly **two minutes** per patient.

288
PER DAY

What the patient produced: a CGM value **every five minutes** — about **3,000 readings** in a ten-day wear — plus continuous heart rate, steps, sleep and every meal. Almost none of it is read.

24/7
WEARABLE

And alongside it: heart rate, step count and sleep, recorded continuously by a watch the patient already owns — **never opened** in the consultation.

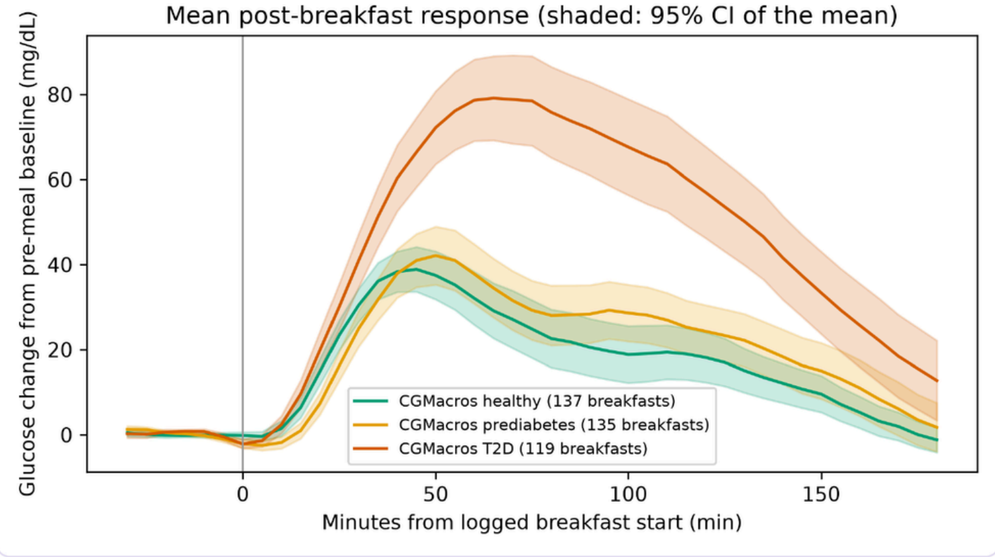
VISIT — NEXT VISIT

~90 days of data, unseen

Same number, different days. Two patients can share an HbA1c and have completely different time in range — one steady, one swinging between highs and lows that average out to something reassuring.

288 = 24 h at 5-minute sampling · Source: README §8

What a three-month average cannot see



Mean post-breakfast response in **CGMacros**, by group, with 95 % confidence bands. Type 2 patients rise about **79 mg/dL** above baseline and stay elevated for hours. An HbA1c shows none of this — not the peak, not the timing, not the recovery.

WHAT EACH SIGNAL CAN TELL YOU

SIGNAL	SAMPLED	CAPTURES	MISSES
HbA1c	Every 3 months	Long-run average exposure	Spikes, lows, timing
Fasting glucose	Per visit	One morning value	Everything after a meal
CGM	Every 5 minutes	The whole day, highs and lows	Why it happened
Wearable	Continuous	Activity, heart rate, sleep	Glucose itself
Meal log	Per meal	The main driver of excursions	Needs the patient to record it

Existing CGM apps **display** these streams. None forecasts them, and none answers “what if the meal were different”.

Source: CGMacros (PhysioNet) · README §8, §9.1

Why a digital twin, and not another dashboard

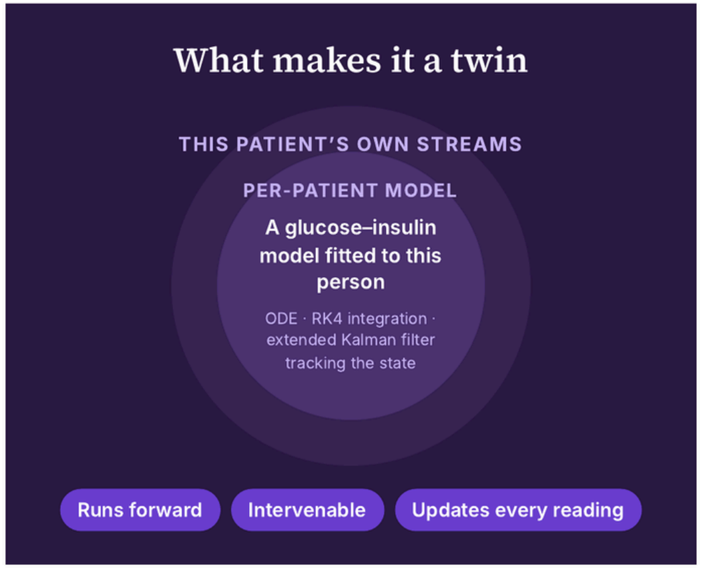
A dashboard answers **what happened**. The decisions taken in a two-minute consultation are about **what happens next** — and what would happen instead. No chart of the past can answer either, because the alternative never occurred.

+120 min how far ahead it forecasts, at **three horizons**, each with a stated range

80 % intervals that are **measured, not assumed** — and reported per subgroup

2 independent what-if engines: a **monotone** learned model and the fitted twin

What makes it a twin



CAPABILITY	CGM APP / DASHBOARD	GLUCOTWIN
Shows the last two weeks	**Yes**	Yes — AGP, time in range, GMI
Forecasts the next two hours	No	+30, +60 and +120 minutes
States how certain it is	No	80 % interval, measured at 80.5 %
Gives the chance of an event	No	A calibrated probability, not an alarm
Answers "what if the meal were different"	No	Two engines, checked for plausibility
Fitted to this patient's physiology	No	Personal basal glucose; state updated live

Every row on the right is measured, not asserted. The interval coverage, the event probabilities and the plausibility checks all have numbers behind them, each computed on patients the model never trained on.

What it is not

- Not an avatar or a 3D body model.
- Not a chatbot placed over the patient's data.
- Not a population risk score applied to an individual.

What the clinician reads

```
Glucose is 338 mg/dL and rising. The model
expects 291 in an hour (80 % range 264 to 312).
Chance of being above 180 in 60 minutes: >99 %.
```

ACTUAL OUTPUT · SIMULATED PATIENT S-017

Where the line is

The clinician remains the decision maker. The system reports what is likely and how certain it is. It **never recommends a dose or a drug**, and it is not a medical device.

Source: README §5, §7 · D-072a · docs/DECISIONS.md

The data: four streams, two real cohorts, nothing private

Every number in this project comes from **open or clearly-labelled synthetic data**. No real patient record is stored in this repository, and none ever was.

WHAT GOES IN

Continuous glucose REAL · OPEN

A CGM value **every five minutes** across 10–14 day wears. The stream the forecast depends on most — remove it and accuracy falls further than for any other input.

Meals REAL + PRESETS

Logged carbohydrate, protein, fat and fibre, plus **16 Indian meal presets** with composition from IFCT-2017. The largest single gain of any added stream.

Wearable REAL + SIMULATED

Fitbit **heart rate and activity** from CGMacros. Sleep stages, heart-rate variability and step count come from our simulator, because no open dataset provides them alongside CGM in type 2 patients.

Health record SIMULATED

Age, sex, BMI, HbA1c, fasting glucose, eGFR, diagnoses, medications and genetic markers, in a **Synthea-compatible schema**.

THE COHORTS

COHORT	PATIENTS	CONTAINS	LICENCE
CGMacros · PhysioNet	45 adults	CGM, Fitbit heart rate and activity, meal macros, labs	CC BY-NC-SA 4.0
Shanghai T1DM/T2DM · figshare	100 T2DM	CGM, meals, labs	CC BY 4.0
Synthetic Indian cohort · ours	100 per setting	CGM, heart rate, steps, sleep stages, HRV, Indian meals, EHR history	Generated · SIMULATED

Throughout the rest of this deck these are called **CGMacros** (45 patients, United States) and **Shanghai T2DM** (100 patients, China). Every result that follows is reported in both.

> **Why two real cohorts, not one.** Different countries, different diets, different clinical practice. One cohort tells you a model works. Two tell you whether it survives being moved — the question that decides whether it could ever be deployed in India.

How the data is handled

145 **real patients** across two open cohorts, two countries

5 min the grid **every cohort is harmonised onto**

0 real patient records **stored in this repository**

- **Fetched, never committed.** Scripts under `data/` download the open datasets and verify checksums. The repository holds code, not patient data.
- **One causal timeline.** Every cohort is resampled onto the same five-minute grid with a shared schema.
- **No feature may look ahead.** Every feature uses only information available at or before the moment of prediction — enforced in code, covered by tests.
- **Patients never straddle a split.** Training, inner-validation, calibration and holdout patients never overlap, so no patient appears on both sides of an evaluation.

THE INDIAN COHORT WE BUILT

100 virtual patients per setting, 10 days each. Meals drawn from 16 Indian presets with carbohydrate from **IFCT 2017**; insulin sensitivity lowered by **0.46 log units** from published clamp data, across four strengths from none to strong. Every screen it touches is labelled SIMULATED.

How the system works, end to end

Four streams in, harmonised onto **one causal timeline**, fused by the model layer, turned into intervals and probabilities by the calibration layer, and served to two interfaces that read the same bundle — so the two can never disagree on a number.

Inside the clinic — raw records never leave

1 · SOURCES

REAL
CGM · 145 patients
Two open cohorts, a value every 5 minutes

REAL
Meals
Macros, plus 16 Indian presets

REAL + SIM
Wearable
Heart rate, activity; sleep and HRV simulated

SIMULATED
Health record
Labs, diagnoses, medications, markers

`data/ · fetched, checksums verified`

2 · HARMONISE

One 5-minute grid
Every cohort on a shared schema, checksums verified

Causal features only
Nothing may use information from after the prediction instant

Patient-grouped splits
No patient on both sides of an evaluation

Quality score
Completeness, recency, plausibility travel with the forecast

`src/data · src/features`

3 · MODEL LAYER

SERVED
Tuned LightGBM
Absolute-error loss, three models averaged; predicts the change at +30/+60/+120 min

Mechanistic twin
Glucose–insulin ODE per patient, RK4 + Kalman filter

Hybrid physics + ML
Twin features, and a GRU with an ODE-consistency loss

Personalisation · federated
Fine-tuned on the patient's days; FedAvg across clinics

`src/models · src/mech · src/hybrid · src/federated`

4 · CALIBRATE

80 % intervals
Split-conformal and CV+ — measured at 80.5 %

Event probabilities
Platt and isotonic for 180, 140 and 70 mg/dL

Robustness
Sensor faults injected, features recomputed

Explainability
SHAP over the fused feature set

`src/uncertainty · src/robustness · src/explain`

5 · OUTPUTS

Forecast + band
To +120 min, with one plain sentence

Adverse-event risk
A calibrated chance, not an alarm

What-if answers
Monotone-constrained: less carbohydrate can never predict a higher peak

Trust surface
Data quality, personalisation depth, limits

`src/whatif · src/dash`

6 · INTERFACES

Web console
Live in a browser. Patient list with triage, forecast, what-if lab, glucose report, health record, visit summary, role-gated access log

Streamlit application
Ten pages including twin state, input tiers, data quality, evidence and a PDF export

One shared bundle
Both read the same numbers

`web/ · app/ · make app`

HOW IT IS TRAINED

LightGBM, one head per horizon, on an **absolute-error loss to match the metric it is scored on**. 16 candidates searched, best three averaged — chosen on inner-validation patients only.

HOW IT IS VALIDATED

Patient-grouped cross-validation with repeats, an inner split for every choice, and a reserved holdout the runner refuses to reuse.

WHAT IT HAS TO BEAT

Persistence, per-patient autoregression, ridge and a **GRU sequence model** — scored on the same windows, with patient-level confidence intervals.

HOW CHOICES WERE MADE

Every protocol, candidate set and decision rule written to **DECISIONS.md before the run that tested it**. Over 70 numbered entries. Nulls published at full size.

Privacy by architecture

Clinics train together through **federated averaging** — only model updates cross the boundary, never records. Data minimisation by design, mapped to India's DPDP Act.

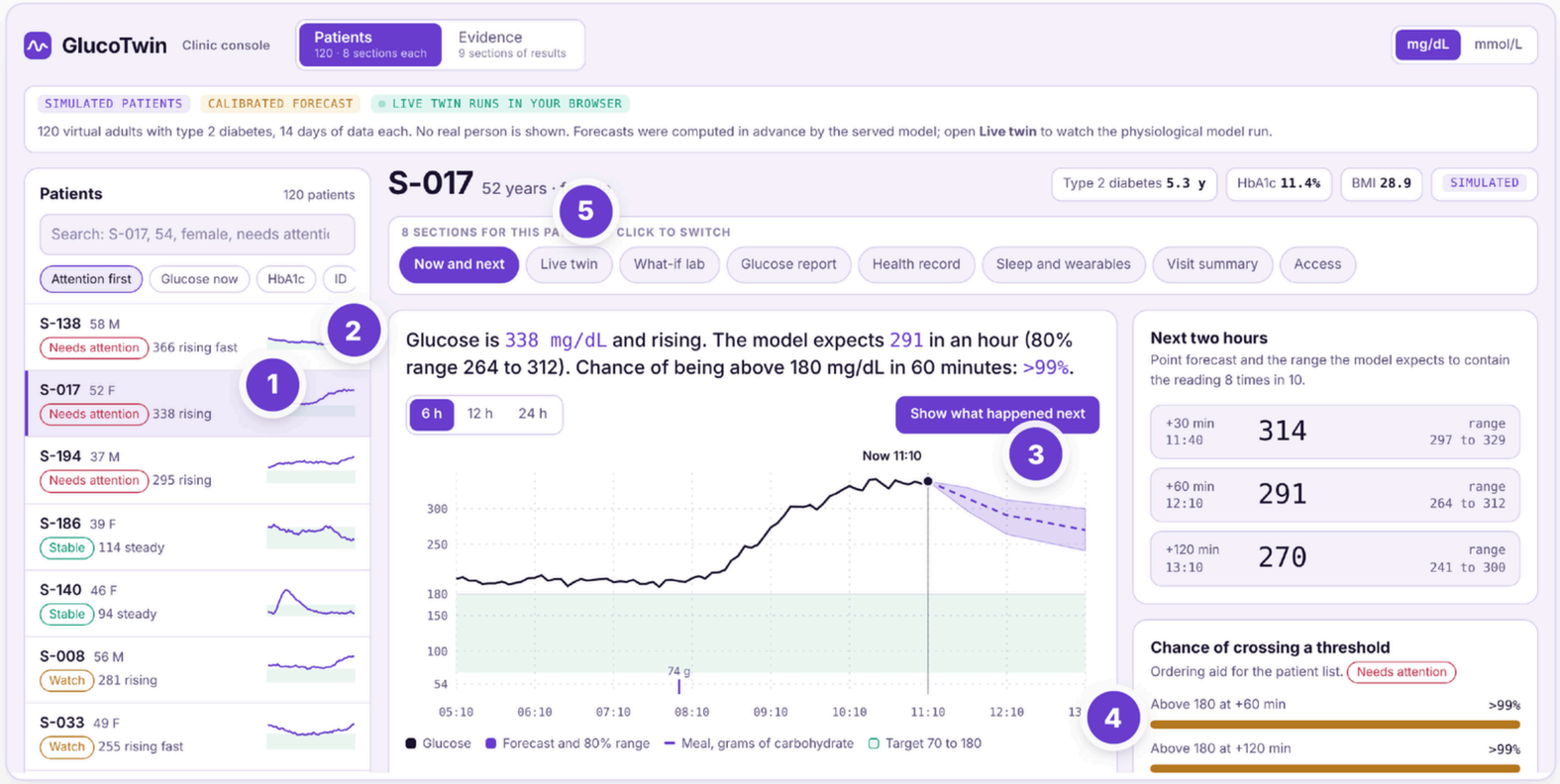
Reproducible, and it runs on a laptop

One **make target per stage**; every figure and table regenerates from source. CI runs lint, tests and a smoke run on every push. Python 3.11, LightGBM, PyTorch, SHAP — **no GPU required**.

Full diagram: docs/architecture.pdf · Source: README §6, §7 · docs/DECISIONS.md

The clinician console

One page, opened per patient, designed for a **two-minute consultation**. It is live in a browser — no install, no sign-in — and it reads the same bundle as the Streamlit application, so the two can never disagree on a number.



WHAT YOU ARE LOOKING AT

1. **A list ordered by who needs attention**
   The day starts with triage, not one patient. Rows are ordered by the **calibrated chance of a dangerous high**, each tagged Needs attention, Watch or Stable.

2. **One sentence a clinician can read instead of the chart**
   Where glucose is now, where it is expected to be, the **80 % range**, and the chance of crossing 180 mg/dL.

3. **The forecast, with its uncertainty drawn**
   History solid, forecast dashed, the 80 % band shaded. **The band widens with the horizon** — that widening is the honest content of the picture.

4. **Calibrated chance, not an alarm**
   A probability the clinic can threshold itself. **40 % means it happens about 40 % of the time** — the calibration is measured, not assumed.

EIGHT TABS PER PATIENT · 5 THE LIVE TWIN RUNS IN YOUR BROWSER

Now and next · **5 · Live twin** · What-if lab · Glucose report · Health record · Sleep and wearables · Visit summary · Access

Live now cleopatrakilledceaser.github.io/SynapseSquad_IITGuwahati

Role views, and logged

A clinician and a care coordinator see **different fields**. Every access is written to a session audit trail in the Access tab — a demonstration of role-based display, not authentication.

Built for the room

A bright consulting room, a laptop or a tablet at 820 px. **mg/dL ⇄ mmol/L**, dark and light themes, body text measured at **15.9:1** contrast, and every glucose state carrying a pattern and a written label — never colour alone.

An evidence section, not just patients

A second view carries **every cohort-level result** behind the forecast — accuracy, interval coverage, alarm burden, what-if plausibility, transfer, federated learning — with nine figures. A judge can check a claim without leaving the page.

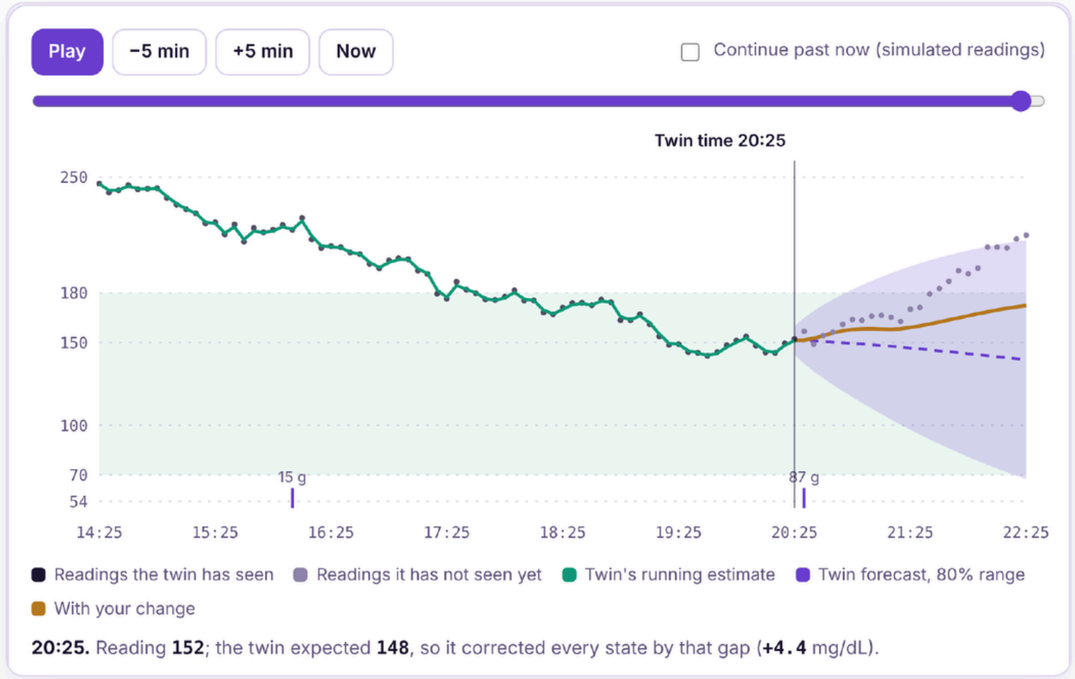
What the demo runs on

The demo runs on **120 simulated patients** with a three-day local calibration. No real person is shown; a SIMULATED badge sits on every screen, and the decision-support disclaimer is on every page.

Live: cleopatrakilledceaser.github.io/SynapseSquad_IITGuwahati · Source: README §10 · D-069, D-070, D-072a

The what-if lab

Before the meal, not after the spike: ask what a different plate or a walk would do. **The physiological twin sizes the change**, and a **monotone model guarantees its direction**.

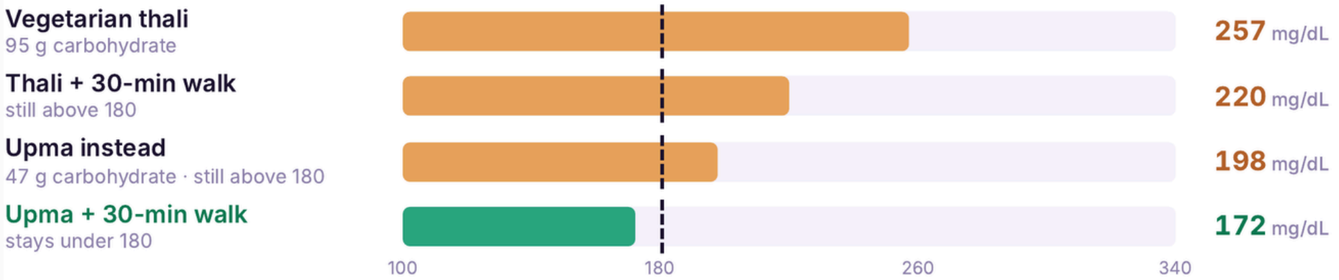


20:25. Reading **152**; the twin expected **148**, so it corrected every state by that gap (**+4.4** mg/dL).

The Live twin tab, running in the browser. Green: the twin's estimate, corrected by every reading. Dashed purple: no change, with its 80 % range. Orange: upma plus a 30-minute walk. Grey dots are readings the twin had not yet seen: in the simulation she ate an 87 g dinner and reached 215 mg/dL.

ONE PATIENT, FOUR PLANS · PREDICTED PEAK IN THE NEXT TWO HOURS

S-151, 53, type 2 diabetes for six years, hypertension, HbA1c 7.6 %. At 20:25 her glucose is **152 mg/dL** and dinner is next.



One change is not enough; the combination is. Simulated patient; the walk's effect is a literature-based assumption (a 20 % smaller three-hour rise), labelled on screen. The browser's numbers are checked against Python in CI (`twin_example_s151.json`).

ENGINE 2 · PHYSIOLOGICAL TWIN

Sizes the change

Carbohydrate response **0.33 mg/dL per g**, inside the pre-registered 0.3–1.3. On real breakfasts eaten twice by the same person it predicts **27.5** of an observed **33.5 mg/dL** on average.

ENGINE 1 · MONOTONE MODEL

Guarantees the direction

Less carbohydrate never predicts a higher peak: **100 %** of carbohydrate checks. Its sizes run low (0.19 mg/dL per g), so they are shown as likely under-estimates.

PHYSIOLOGICAL PLAUSIBILITY CHECKS · TEST PATIENTS (CGMACROS)

CHECK	TWIN	MONOTONE	UNCONSTRAINED	WHAT IT TESTS
Carbohydrate direction	0.976	1.000	0.275	More carbohydrate never lowers the peak
Walk direction	1.000*	0.999	0.679	A walk does not raise the peak
Identity · causality · bounds	0.958	1.000	1.000	No edit, no change; no effect before the meal; plausible range
Dose window	0.965	1.000	0.983	The effect lands in a physiological window
Dose response, mg/dL per g	0.33	0.19	0.20	Inside the pre-registered 0.3–1.3 range

Each engine where it measured best

The twin gets the **size** of a meal right on average; the monotone model never gets the **direction** wrong. The console shows both: the Live twin and the What-if lab tabs.

Where the evidence stops

Both engines match the **average** effect better than any single meal (per-pair r ≤ 0.21). The walk effect is assumed, and the twin does not model insulin doses or sulfonylureas. *Walk direction uses the one-sided activity term added in D-056, labelled post-hoc.

Source: README §9.3 · D-050, D-056, D-072a, D-075 · reports/metrics/whatif_plausibility.json, whatif_matched.json, twin_example_s151.json

How accurate is the forecast?

Every model is measured against the same patients, the same splits and the same honest baselines. **Mean absolute error in mg/dL — lower is better.**

CGMACROS · 45 PATIENTS · HELD-OUT TEST PATIENTS · 95 % PATIENT-BOOTSTRAP INTERVALS

MODEL	+30 MIN	+60 MIN	+120 MIN
Persistence Repeat the last reading — the clinical default	13.4	19.5	26.9
Per-patient autoregression A model fitted to that one patient's own series	12.2	17.4	21.8
LightGBM, CGM only Learned, but glucose history and nothing else	12.3	17.7	22.0
LightGBM, CGM + wearable + meals All four streams fused	11.6	16.8	21.2
Served model · tuned (D-072a) Trained on absolute error, the metric it is scored on — this is what both dashboards run	**10.88**	**15.70**	**19.96**

THE GAP AT +60 MINUTES, DRAWN TO SCALE

A 19 % cut against the clinical default, and the gain is where it matters. In the three hours after a meal — the window a clinician is actually deciding in — fusing meals into the model is worth 1.57 mg/dL, against 0.93 averaged over the whole day.

Reported per subgroup

Error and interval coverage are broken out by **HbA1c band, age and sex**, as well as pooled — so a model that works on average but fails for one group cannot hide inside a single number.

Robustness tested, not assumed

Sensor faults are **injected into the test data** and every causal feature recomputed. A data-quality score — completeness, recency, plausibility, rate of change — travels with each forecast.

The nulls are published too

Wearable heart rate and activity, added on their own, move error by **+0.04 mg/dL with an interval that includes zero**. It is in the table above and in the README, not left out.

Source: README §9.1 · D-010, D-030, D-072a · reports/metrics/forecast_baselines.json, forecast_served.json

Does it hold in a second cohort?

100 **Shanghai T2DM patients** — a different country, diet and clinical practice

15.51 mg/dL at +60 min, the **served model**, replicating the first cohort

80.5 % of true values fall inside the **80 % band** the model promised

HOW THE COMPARISON IS KEPT HONEST

- **The baselines are real, not strawmen.** Persistence is genuinely hard to beat over thirty minutes of CGM, and the autoregression is fitted to each patient individually.
- **Intervals are resampled over patients, not readings.** Two readings from one person are not two independent facts, so every difference carries a paired patient bootstrap.
- **Patients never straddle a split.** Training, inner-validation, calibration and holdout patients never overlap.
- **Two cohorts, not one.** Every headline number is reported in both, so nothing rests on a single dataset of 45 people.
- **The holdout was opened once.** Tuning saw inner-validation patients only, and the rule for adopting the tuned model was written before the run.

WHAT THE TUNING BOUGHT

Training on absolute error and searching 16 candidates moved every number the right way at once: **error down and the 80 % band narrower** at the same promised coverage, in both cohorts at all three horizons. At +60 the band tightened from **50.6 to 47.8 mg/dL** while coverage held at 80.5 %.

What is each data stream actually worth?

Remove one group of features, **retrain the whole model**, and measure how much the error rises. That rise is what the stream was carrying — paired over the same patients, with 95 % intervals.

CGMACROS · 45 PATIENTS · +60 MIN · MG/DL OF ERROR *ADDED* WHEN THE STREAM IS TAKEN AWAY

STREAM REMOVED	ACROSS THE WHOLE DAY	IN THE 3 H AFTER A MEAL	IN THE RUN-UP TO A HIGH
Meals and logged carbohydrate Carbohydrate, protein, fat, fibre and meal timing	**+0.92** 0.66 to 1.20	**+1.56** 1.08 to 2.06	**+2.01** 1.52 to 2.53
HbA1c and fasting glucose Already in every clinic's record	**+0.61** 0.07 to 1.30	**+0.90** 0.20 to 1.64	**+0.80** 0.29 to 1.42
Wearable heart rate and activity Fitbit, measured — not simulated	**+0.12** 0.05 to 0.22	**+0.12** 0.00 to 0.25	**+0.02** −0.16 to 0.21

AND FOUR GROUPS THAT CARRY NOTHING · INTERVAL INCLUDES ZERO IN EVERY CASE

+0.04 **Overnight rest proxy** −0.02 to 0.10	−0.01 **Age, sex, BMI** −0.11 to 0.09	−0.01 **Fasting insulin** −0.13 to 0.10	−0.06 **Lipid panel** −0.19 to 0.07

THE SAME TWO STREAMS, DRAWN TO ONE SCALE · MG/DL ADDED WHEN REMOVED

One stream grows as the decision gets closer. The other disappears.

Removed and retrained, not ranked

This is a **leave-one-group-out ablation**, not a feature-importance score read off a fitted model. Importance scores split credit arbitrarily between correlated features; retraining without the stream does not.

Detectable is not the same as useful

Wearables cost **0.12 mg/dL** when removed — an interval that excludes zero. Added to CGM alone they are worth **0.04**, and that interval includes zero. Both numbers are published; neither is worth a claim.

Repeated in the second cohort

Shanghai T2DM carries no wearable or macro logging. Of **seven feature groups beyond CGM, seven are nulls** — not one is distinguishable from zero. We report the empty result as carefully as the positive one.

Source: README §9.1 · D-030, D-031 · reports/metrics/fusion_cgmacros.json, fusion_shanghai_t2dm.json

What this changes

+2.01 mg/dL the **meal log** is worth in the run-up to a high — the decision that matters

4 of 7 feature groups in CGMacros carry **nothing distinguishable from zero**

0.12 mg/dL from the **wearable** — real, measured, and far too small to sell

THREE DECISIONS THIS FORCED

- **Spend the design effort on the meal log.** It is the one input worth real accuracy — which is why the console offers 16 Indian presets and asks for katoris and plates rather than grams.
- **Do not sell the wearable.** A product claiming the watch drives the forecast would be claiming something we measured and did not find.
- **Collect meals before collecting sensors.** On this data a clinic that adds a meal log gains roughly **eight times** what one adding a wrist tracker gains.
- **Keep HbA1c and fasting glucose in the model.** Second most valuable, and free — they are already in the chart before the patient walks in.

SO WE BUILT THE MEAL LOG PROPERLY

16 Indian presets with composition from IFCT 2017, asked for in **katoris, rotis and plates** rather than grams, each one carrying carbohydrate, protein, fat and fibre. The highest-value input in the model is the one a patient can enter in a single tap.

Four model families. The simplest one won.

We did not pick an architecture and defend it. We built a gradient-boosted forecaster, a mechanistic ODE twin, a sequence model and a physics-constrained hybrid, scored them against each other on the same patients, and **shipped whichever won on inner-validation patients**. The deep models did not win — and that is on the slide.

THE RULE, WRITTEN FIRST

Serve the model that wins on **inner-validation patients only**. The holdout is opened once, afterwards. Recorded as D-072 before any of these runs.

SERVED — 15.70

Tuned gradient boosting

LightGBM · one head per horizon

- Predicts the **change** in glucose, not the level — the level is already known
- Trained on **absolute-error loss**, the same metric it is scored on
- **16 candidates searched**, the best three averaged
- A second copy carries a **monotonicity constraint** for the what-if lab

Shipped. Best MAE, tightest intervals, and fast enough to answer a what-if while the patient is still in the room.

SECOND ENGINE — 18.11

Mechanistic digital twin

Glucose–insulin ODE · RK4 · extended Kalman filter

- A **physiological state** per patient — glucose, insulin, gut absorption
- Parameters **fitted to that patient's own days**, with identifiability checked
- The filter keeps the state aligned with each new CGM reading

Shipped for state and what-if. It runs live in the console, corrected by every reading, and sizes meal changes inside the literature range.

TESTED, NOT SHIPPED — 18.00

Sequence model

GRU over 6 h · FiLM conditioning

- Reads the raw six-hour window rather than engineered summaries
- Meals, time of day and patient context injected through **FiLM layers**
- Same splits, same seeds, same loss as the boosted model

Indistinguishable, not better. +0.84 mg/dL against boosting, interval −0.34 to +2.17. No reason to ship the heavier model.

NEGATIVE RESULT — 20.75

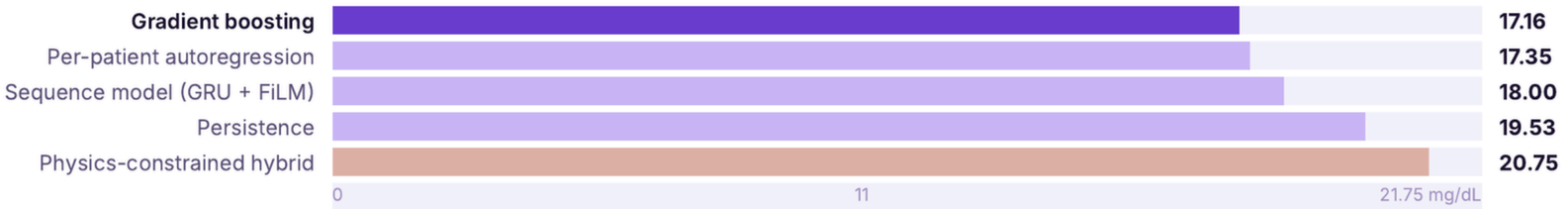
Physics-constrained hybrid

GRU + twin features + ODE-consistency loss

- Twin-derived states handed to the network as extra features
- A loss term penalising departure from the **ODE's own trajectory**
- The obvious idea, and the one we most wanted to work

It made things worse. +3.59 mg/dL against boosting, interval +2.51 to +4.76. Published rather than dropped.

LIKE FOR LIKE · CGMACROS · +60 MIN · SAME PATIENTS, SAME SPLIT, CROSS-VALIDATED



One evaluation, one split, one seed policy. The mechanistic twin is scored in its own run (18.11 at +60) because it fits parameters per patient. The served figure of 15.70 is the tuned model on the held-out test patients, reported separately.

Why the deep model losing is the point

On **45 patients and roughly 40,000 windows**, a sequence model has nothing to learn that engineered causal features have not already given it. Reporting that is more useful to the field than shipping a heavier model for the sake of the word *neural* — and it is why the console runs in a browser with no server behind it.

THE SERVED MODEL AT A GLANCE

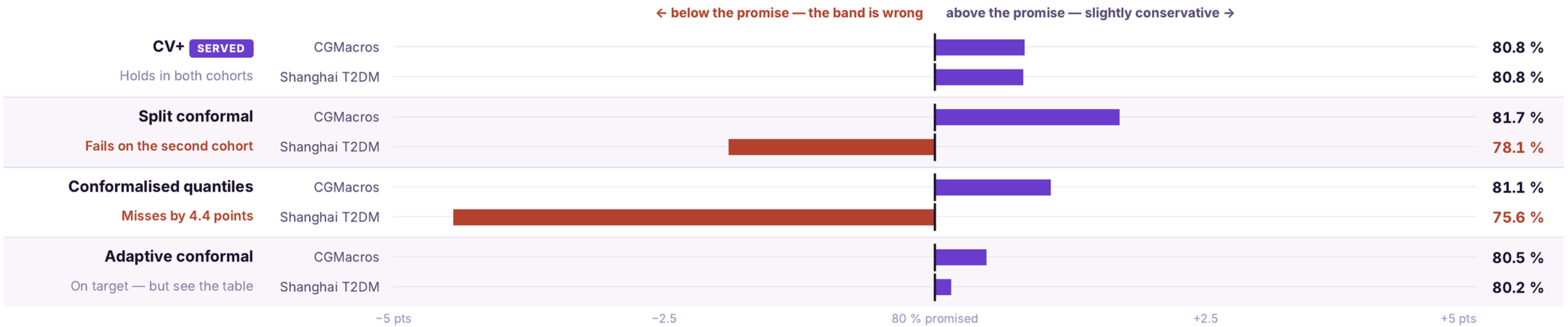
INPUT WINDOW	TARGET	HEADS	LOSS	SELECTION	RUNS IN
6 hours	**Δ glucose**	**3 × 2**	**L1**	**16 → 3**	**the browser**
of causal history, nothing after the prediction instant	the change, at +30, +60 and +120 min	one model per horizon, per cohort	absolute error — the metric it is judged on	candidates searched, best three averaged	no server, no install, no sign-in

Source: README §9.1, §9.6 · D-040, D-041, D-060, D-072 · reports/metrics/hybrid_neural.json, mech_cgmacros.json, fusion_cgmacros.json

Can you trust the interval?

A forecast without an interval is a guess with a decimal point on it. So we built four conformal methods, and judged them on one question only: **does an 80 % band actually contain the truth 80 % of the time, in both cohorts, at every horizon?** Three of the four fail somewhere. The slide shows where.

DOES THE 80 % BAND ACTUALLY HOLD 80 % OF THE TIME? · AT +120 MIN, THE HARDEST HORIZON



WHAT EACH METHOD COSTS · MEDIAN 80 % BAND WIDTH AT +120 MIN, CGMACROS

METHOD	BAND WIDTH	INTERVALS THAT ARE INFINITE	WHAT THAT MEANS IN THE CONSOLE
CV+ — served	**61.0 mg/dL**	**none**	Every forecast gets a finite, usable range. Holds its promise in both cohorts.
Split conformal	64.3 mg/dL	none	Wider than CV+ *and* less reliable — strictly worse, so it is not served.
Conformalised quantiles (CQR)	57.4 mg/dL	none	The tightest-looking bands on the slide. They are tight because they are too small.
Adaptive conformal	47.1 mg/dL	**34.6 % · 43.6 %**	It reaches 80 % by answering “anywhere” for a third to a half of predictions. Honest, and useless at the bedside.

Source: README §9.2 · D-051, D-052, D-072a · reports/metrics/uncertainty.json (80 % level, served configuration)

WHAT COVERAGE MEANS

A band is a promise

Coverage is the only number that tells a clinician whether the uncertainty on the screen means anything. **We picked the method that keeps the promise**, not the one with the prettiest bands.

THE TRADE-OFF, NAMED

Narrow is not better

Three of these four methods produce narrower intervals than the one we serve. Two get there by **undercovering**, one by **refusing to answer**. Width without coverage is decoration.

CHECKED BY SUBGROUP

Where it holds, and where it does not

Every HbA1c band of the all-type-2 Shanghai cohort: **79.5 to 83.7 %**. CGMacros' 14 type 2 patients: **73.8 %**, calibrated on a mostly non-diabetic pool. Group-wise calibration is the next fix.

Warning before it happens

The competition asks for a two-hour warning of an adverse event. GlucoTwin gives one as a **calibrated chance** — a number the clinic can act on and set its own threshold against — backed by eighteen dedicated classifiers and measured in both cohorts.

RANKING WHO IS AT RISK · AUROC, HELD-OUT TEST PATIENTS

Still ranks risk well two hours out

EVENT · COHORT	+30 MIN	+60 MIN	+120 MIN
Above 180 · CGMacros	0.91	0.83	**0.77**
Above 180 · Shanghai	0.96	0.89	**0.82**
Above 140 · CGMacros	0.85	0.77	**0.73**
Above 140 · Shanghai	0.92	0.85	**0.80**
Below 70 · CGMacros	0.95	0.88	**0.78**
Below 70 · Shanghai	0.97	0.91	**0.81**

Every event clears **0.90 at thirty minutes** in at least one cohort and holds above **0.73 at two hours** everywhere — the competition's own case. Each figure carries a 95 % interval resampled over patients.

TURNING A SCORE INTO A CHANCE · CALIBRATION ERROR AT +60 MIN

A 40 % chance happens about 40 % of the time

0.025

Platt — served. Worst case across both cohorts and all three events

0.018

Isotonic. Also fitted and reported; Platt is the steadier choice at this cohort size

A raw model score is not a probability. We fit a calibration map, then measure how far the stated chance sits from the rate actually observed. **At most 2.5 percentage points out** — which is what lets the console print a number a clinician can set a threshold against.

STEP 1
Train
Training patients only

STEP 2
Calibrate
A separate group of patients, never trained on

STEP 3
Measure
Held-out patients, seen by neither step

AT THE SAME ALERT LOAD · SHANGHAI T2DM, GLUCOSE ABOVE 180

More episodes caught, same number of alerts

All three methods are tuned to raise the **same number of alerts per patient per day**, so the comparison is like for like. Against the alert a clinic runs today, that is **24 more hyperglycaemia episodes caught in every hundred**, at no extra load on the nurse.

The **full threshold sweep ships in the console**, so a clinic chooses its own point on the curve rather than inheriting ours.

WHAT SITS BEHIND THE NUMBER

How the warning is built

1. **Eighteen dedicated classifiers.** One for each of three events at each of three horizons, in each cohort — trained separately from the forecaster, because ranking who is at risk and predicting a value are different jobs.
2. **Three scorers compared for every one of them:** the dedicated classifier, the forecast value itself, and the current reading. The classifier is served because it won on inner-validation patients.
3. **Two calibrators fitted and compared** — Platt and isotonic — on patients used for neither training nor testing.
4. **The whole threshold sweep computed**, not a single operating point, so the clinic owns the trade-off and can move it.
5. **Reported per subgroup as well as pooled**, with patient bootstrap intervals on every figure.

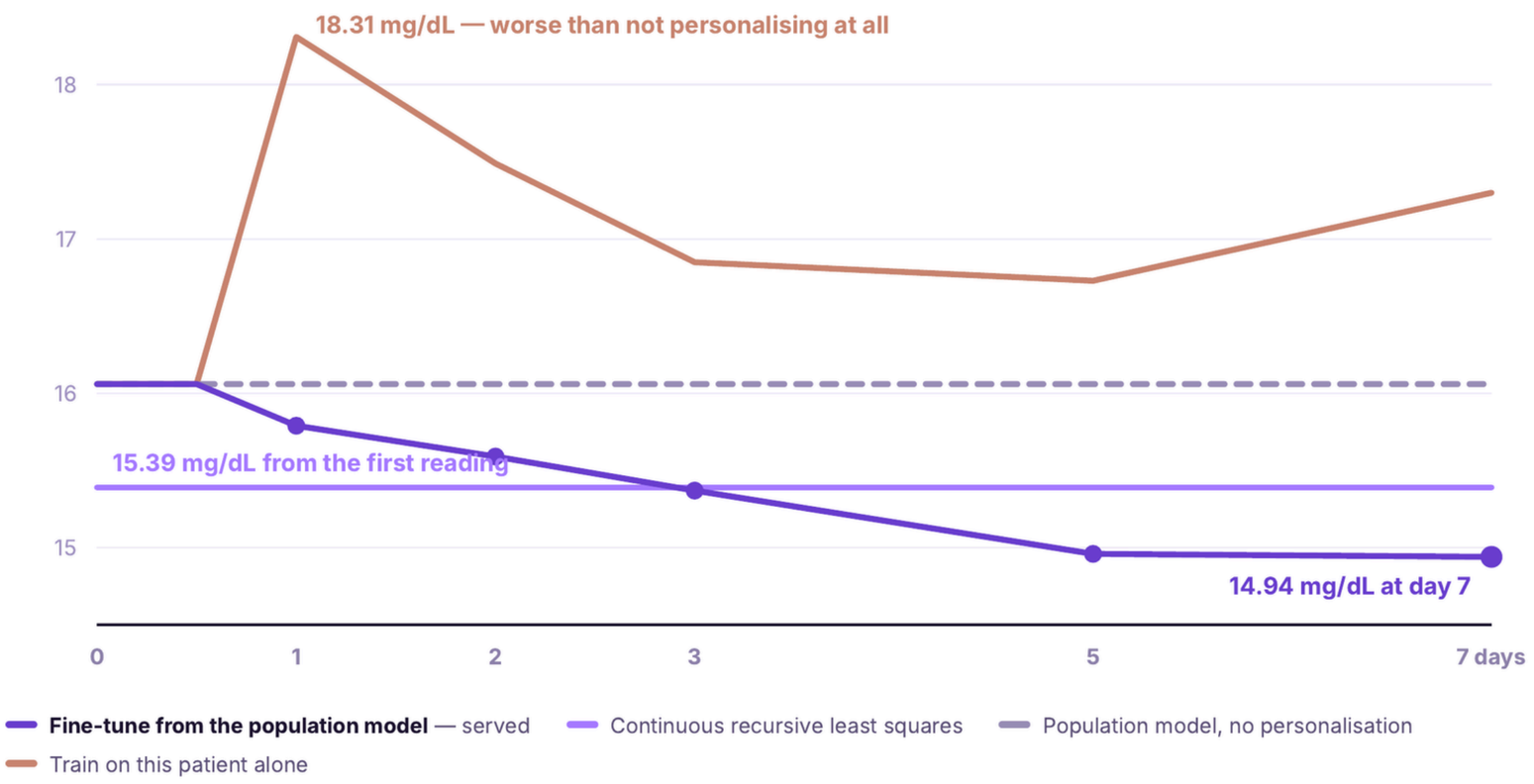
Each of these is a **pre-registered decision** in **docs/DECISIONS.md** — written before the run, not chosen after seeing the results.

Source: README §9.2 · D-020, D-021, D-053 · reports/metrics/baselines_event_onset.json, baselines_cgmacros.json, baselines_shanghai_t2dm.json

What is a patient's own history worth?

A twin should get better at *you* the longer it watches you. We tested **six ways of personalising** against a population model, over seven days of a patient's own readings, and measured every one of them on the same held-out patients.

CGMACROS · MAE AT +60 MIN, MG/DL · LOWER IS BETTER · DAYS OF THAT PATIENT'S OWN DATA



Replicated in Shanghai T2DM, 100 patients: fine-tuning falls **15.68 → 14.65 mg/dL** by day seven, continuous RLS gives **14.99** from the first reading, and training on the patient alone rises to **19.18**. The same three findings, a different country and diet.

The population prior does the heavy lifting

A model trained on one patient's own days alone reaches **18.31 mg/dL** against **16.06** for no personalisation at all, and never catches up within a week. Starting from 145 patients is what makes one patient's data useful.

Fine-tuning keeps paying, day after day

Adapting the population model to the patient falls from **16.06 to 14.94 mg/dL** by day seven — a **7 % cut**, still improving at the end of the window. In the run-up to a high, where the error is largest, the gain is bigger still: **23.41 to 21.87**.

One method pays from day zero

Continuous recursive least squares corrects the forecast **online, as each reading arrives — 15.39 mg/dL with no warm-up window at all**. A new patient benefits on their first morning, before any history exists.

The console says which one you are seeing

Every forecast carries a **personalisation depth** — how many days of that patient's own data sit behind it — so a clinician knows whether they are looking at a population estimate or a twin that has been watching this person for a week.

SIX STRATEGIES TESTED · MAE AT +60 MIN AFTER THREE DAYS OF THE PATIENT'S OWN DATA

15.37	**15.39**	**15.45**	**15.47**	**15.92**	**16.85**
Fine-tune	**Continuous RLS**	**Shrink to population**	**Recursive least squares**	**Residual model**	**Train from scratch**
Continue training the population model on this patient	Online correction, no training window needed	Blend a personal fit toward the population mean	Refit a linear correction over the window	Learn this patient's error on top of the population model	Use only this patient's own readings

Taking the model to a new country

A model that only works on the patients it was trained on is a research result, not a product. So we took each cohort's model to the other country, **measured exactly what the move cost**, and then built a synthetic Indian cohort to test the move that actually matters.

THE SAME MODEL, USED ON A COUNTRY IT WAS NOT TRAINED ON · MAE AT +60 MIN, MG/DL · LOWER IS BETTER

US CGMacros (45 patients) → **CN Shanghai T2DM** (100 patients)

Repeat the last reading	**19.50** mg/dL
Taken across unchanged	**19.37** mg/dL
Trained on both cohorts pooled	**15.98** mg/dL
Trained on local patients	**15.95** mg/dL

Taken across unchanged, the model does no better than repeating the last reading. Trained on local patients it is **3.4 mg/dL better** — and pooling both countries reaches the same place. Both are scored on the **same held-out patients**, on a shared 15-minute grid, so the only thing that changes is where the training data came from.

CN Shanghai T2DM (100 patients) → **US CGMacros** (45 patients)

Repeat the last reading	**19.31** mg/dL
Taken across unchanged	**20.08** mg/dL
Trained on both cohorts pooled	**17.85** mg/dL
Trained on local patients	**17.49** mg/dL

The same result in the other direction, on a cohort less than half the size: **2.6 mg/dL** separates a model that has seen local patients from one that has not. Running the move **both ways** means the finding cannot be explained by one cohort simply being larger or cleaner than the other.

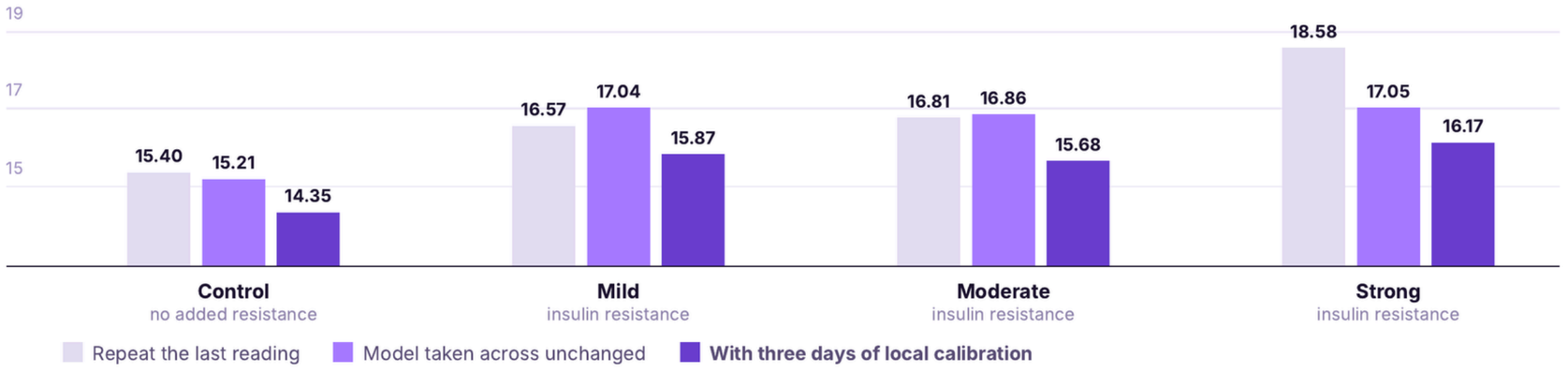
WHAT THE MOVE COSTS

3.4

mg/dL behind a locally trained model · 95 % CI 2.9 to 4.0

- **Pooling the data recovers it entirely** — 15.98 against 15.95 trained in-cohort.
- **Three days of the new patient's own data closes part of it** — a further −0.83 mg/dL, CI −1.19 to −0.50.
- So the recipe for a new country is **train or calibrate locally** — which is exactly what the console does on arrival.
- Both directions were run, so the finding does not rest on **one cohort being larger than the other**.

THE MOVE THAT MATTERS · SYNTHETIC INDIAN COHORT, 100 VIRTUAL PATIENTS PER SETTING · MAE AT +60 MIN, MG/DL



How the cohort was built

100 virtual patients per setting over 10 days. Meals from **16 Indian presets** with carbohydrate from **IFCT 2017**; insulin sensitivity lowered by **0.46 log units** from published clamp data, across four strengths.

What it shows

With three days of local calibration the model **beats repeating the last reading in every setting** — by 0.7 to 2.4 mg/dL. The harder the population, the larger the gain: at strong resistance it is **18.58 against 16.17**.

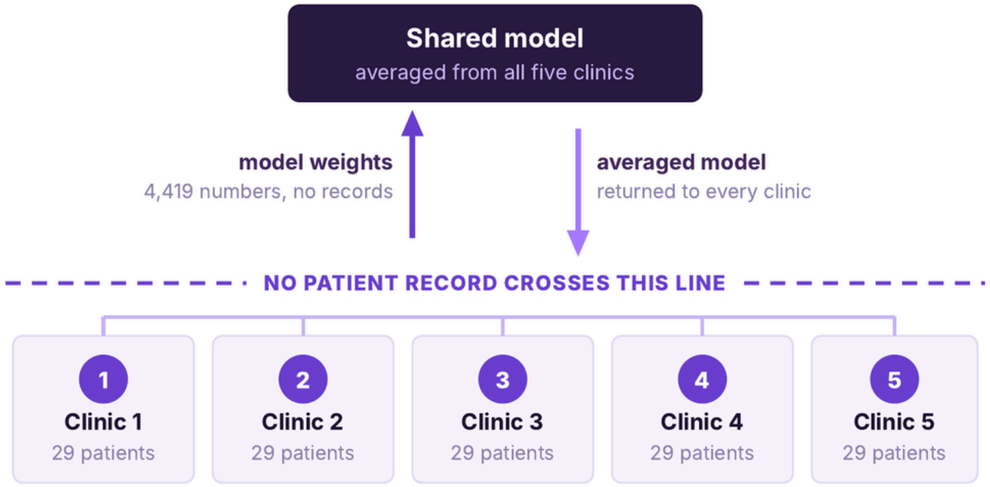
The synthetic cohort stress-tests and demonstrates — it never trains the served model. We checked: adding these patients to training changed nothing worth having, so they stay out of it and every screen they touch is labelled SIMULATED.

Source: README §9.4(a), §9.4(b) · D-063, D-064, D-065 · reports/metrics/transfer.json, india_*.json · IFCT 2017 · JCEM 2001 clamp data

Learning across clinics without moving patient data

An Indian hospital network cannot post its records to a central server, and should not have to. We built the federated version and measured what it costs: **a shared model trained across five clinics, with no patient record ever leaving the building it was recorded in.**

HOW ONE TRAINING ROUND WORKS · SIMULATED CLINICS



STEP 1
Train locally
Each clinic fits the shared model on its own patients

STEP 2
Send weights
Only the 4,419 model parameters leave the building

STEP 3
Average
FedAvg combines them into one shared model

STEP 4
Repeat
40 rounds, and every clinic ends with the same model

WHAT FEDERATION COSTS · 145 OPEN-COHORT PATIENTS GROUPED INTO 5 SIMULATED CLINICS · MAE AT +60 MIN, MG/DL

Clinics with similar patients the straightforward case

Clinics split by HbA1c the realistic case — each clinic sees a different kind of patient

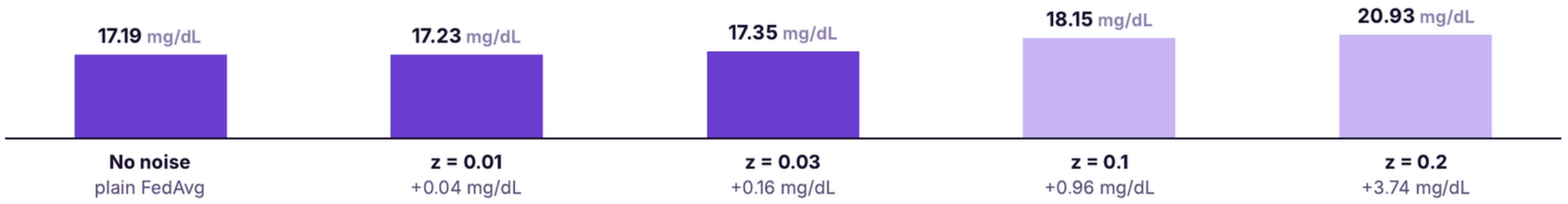
Federation costs 0.09 to 0.13 mg/dL against pooling every record in one place — well inside the confidence intervals, in both layouts. Against a clinic working alone it is worth **2.2 to 2.4 mg/dL**. FedProx was run alongside FedAvg and landed in the same place (17.21 and 17.02), so the simpler algorithm is the one we keep.

40
training rounds between the clinics and the shared model

4,419
numbers sent per clinic per round — the model, nothing else

0
patient records transmitted at any point in the run

WE ALSO MEASURED THE PRICE OF FORMAL PRIVACY NOISE · FEDAVG, SIMILAR CLINICS · MAE AT +60 MIN, MG/DL



WHAT THE CURVE IS FOR

The noise dial, with its price measured

Clipping and Gaussian noise are built in and the **whole trade-off is measured**, not asserted. At **145 patients and 4,419 parameters** the noise needed for a small formal ε costs more accuracy than the model can spare — the ε for every setting is computed and published — so the guarantee this design relies on is **structural: records never leave the clinic**. Data handling is mapped to **India's DPDP Act** in docs/limitations.md.

Would a sleep tracker improve the forecast?

No open dataset pairs real CGM with real sleep staging in type 2 diabetes, so the question cannot be answered from existing data. We built a simulator that **injects a sleep effect of known size**, and asked how big a study would have to be before that effect showed up in a forecast.

9,000
patient-days tested
300 virtual patients × 30 days · 777,000 five-minute readings

5×
the published effect size
sleep's effect on insulin sensitivity, inflated five-fold

At that scale, with the effect exaggerated that far, adding sleep stages and heart-rate variability to the model changes the forecast by **at most 0.09 mg/dL**, on intervals that include zero. The effect is unmistakably present in the data — inflating it moves mean glucose from **161.7 to 163.3 mg/dL** and time above 180 from 27.5 % to 28.4 %. **The model simply does not need the extra channel to see it.**

METHOD

A test real data cannot run

1. **Inject a known effect.** The simulator lowers insulin sensitivity after poor sleep by an amount we choose, so the ground truth is exact — impossible to arrange with real patients.
2. **Vary its strength.** None, the published value, three times and five times it.
3. **Vary the feature set.** Three models: with the sleep channel, without it, and with overnight summaries added.
4. **Vary the study size.** 1,000 patient-days first, then 9,000 — larger than any open CGM cohort that exists.

THE CHAIN THE SIMULATOR BUILDS

A short night
↓
Insulin sensitivity falls by a set amount
↓
Glucose runs higher the next day
↓
The CGM records it

So the sleep channel has to add something this chain has **not already written into the glucose trace**.

Everything on this slide is **SIMULATED** and labelled as such wherever it appears in the app. It tests a study design, not a patient.

RESULT · MAE AT +60 MIN, MG/DL · LOWER IS BETTER

The extra channel never pays

STRENGTH OF THE INJECTED SLEEP EFFECT	WITHOUT SLEEP	+ SLEEP & HRV	+ OVERNIGHT
None	14.29	14.29	14.38
The published value	14.27	14.35	14.31
Three times it	14.41	14.41	14.44
Five times it	14.40	14.39	14.46
At 9,000 patient-days	**12.84**	**12.89**	**12.86**

55 features
Glucose, meals, activity, time of day

76 features
The same, plus sleep stages and HRV

81 features
Plus overnight rest summaries

THE EFFECT REALLY IS IN THE DATA

INJECTED STRENGTH	MEAN GLUCOSE	TIME ABOVE 180
None	161.7 mg/dL	27.5 %
The published value	162.0 mg/dL	27.7 %
Three times it	162.7 mg/dL	28.0 %
Five times it	163.3 mg/dL	28.4 %

Every row is paired over the same patients with 95 % intervals. **No column separates from any other by more than 0.09 mg/dL** at any effect strength or study size — while the same models beat repeating the last reading by **4.9 mg/dL**.

WHAT THIS CONTRIBUTES

The CGM has already seen the night

- Glucose is **downstream of sleep**. By the time a reading exists, last night's effect is already inside it — so a wearable channel has to add something the glucose trace has not recorded, and here it does not.
- This is **a bound, not a shrug**: under the simulator's physiology — itself built from published clamp data — a study designed to find this effect in a forecast needs to be **larger than 9,000 patient-days**, larger than any open CGM cohort in existence.
- That is a **sample-size number anyone planning this work can use**, and we have not found it published anywhere.
- The wearable channels that **do** exist in a real cohort — Fitbit heart rate and activity — were tested separately on real patients and are worth **0.12 mg/dL** (slide 9). Both answers were measured; neither was assumed.
- Together they give a clear engineering instruction: **spend on the meal log and the CGM**, and treat the wrist as a comfort feature until a cohort exists that can prove otherwise.

It is also why GlucoTwin **makes no sleep-driven claim**. The channel is collected, displayed and available — and the model is built on what we could measure.

Source: README §9.5 · D-066, D-067 · reports/metrics/india_sleep_power.json, india_sleep_scale.json · SIMULATED cohort · effect size from published sleep-restriction clamp studies

Every choice, written down before the run

A model can always be made to look good by picking the comparison once you have seen the answer. So every such choice here was recorded first, in a plain-text decision log: **75 numbered entries, D-001 to D-075**, each saying what would be done, why, and what would count as it having worked.

FOUR OF THE SEVENTY-FIVE ENTRIES, QUOTED FROM DOCS/DECISIONS.MD

Each entry carries its **number**, what was decided, **when it was written** relative to the work it governs, and the reasoning behind it.

D-050 — The what-if engine: models, edit semantics and bands

WRITTEN BEFORE ANY WHAT-IF CODE EXISTED

Fixed the two engines, the monotonicity constraint, the fold protocol and the seeds in advance — and recorded the scope limit up front:

> *"Shanghai is excluded from meal what-ifs: its model has no meal inputs and its meal log has no macros. Stated in the report, not dropped silently."*

D-072 — Serve the tuned forecaster, and re-run calibration for it

WRITTEN BEFORE THE MODEL WAS SCORED

Set the adoption rule and the **75 % coverage floor** before the tuned model was ever scored, so the decision to ship it could not be made by looking at the answer. Coverage came back at 80.3 % or better at every horizon, in both cohorts.

D-010 — Recover the real CGM readings before using any of them

PREPARING THE DATA

CGMacros ships one-minute rows with both glucose columns **linearly interpolated**. A linear interpolant bends only at a genuine reading, so a change in slope marks a real one — **99.98 % of 11,915 straight spans** have integer values at every interior point. Anything bridged across a gap longer than 30 minutes is treated as missing; the raw rows hide gaps of up to **9.4 days** in one participant.

D-002 — Stratify the splits by diabetes status

SPLITTING THE PATIENTS

With cohorts of a few dozen patients, an unstratified split can leave a class out of the test set entirely. Allocation uses largest-remainder rounding inside each stratum, with **at least one training patient guaranteed per stratum**.

WHAT THE RECORD CONTAINS

75 — **entries in the decision log** — 71 set the protocol *before* the run; the rest are labelled records

212 — **unit tests**, run by `make test`, covering the causal-feature and split rules

33 of 33 — result files carry the **commit, the data hash and the seed** that produced them

49 — **figures** regenerated from those same files, never redrawn by hand

ANYONE CAN CHECK THIS

Every number on every slide traces to a file

The source line at the foot of each slide names the decision and the `reports/metrics/*.json` it came from. Clone the repository, run `make test` and `make app-data`, and the same numbers come back — **the repository holds code, not patient data**, and the open cohorts are fetched with their checksums verified.

THE RULE EVERY RESULT ON THIS DECK WENT THROUGH

1. **Write the decision** — What will be run, why, and **what would count as it having worked** — before the code exists.
2. **Fix the splits** — Patients are assigned once. Training, inner-validation, calibration and holdout **never overlap**.
3. **Run it once** — Choices are made on **inner-validation patients only**. The holdout stays closed.
4. **Record the provenance** — Commit, data hash, seed and configuration are written **into the result file itself**.
5. **Publish what came out** — Every figure in the deck is **regenerated from those files**, so the slide and the repository cannot drift apart.

Source: docs/DECISIONS.md (D-001 to D-075) · reports/metrics/*.json · tests/ · Makefile · README §16 Reproduce

What we built, measured, and did not ship

Picking the right model is easy once you have built the alternatives and scored them against each other. **Every option below was implemented in full, run on the same patients and the same splits, and set aside on a number** — not on a hunch. All of it is in the repository.

4	4	6	16	
model families built and scored against each other	**interval methods** compared on coverage, in both cohorts	**personalisation strategies** run over seven days each	**hyperparameter candidates** searched for the served model	One of each is served. **Everything else is still in the repository, beside the number that ruled it out** — so the choice can be re-run and checked rather than taken on trust.

WHAT WE BUILT	WHAT THE MEASUREMENT SAID	WHAT WE SERVE INSTEAD
A sequence model over the raw window GRU with FiLM conditioning on meals and context	**+0.84 mg/dL** against gradient boosting, interval −0.34 to +2.17 — indistinguishable, not better	The **boosted model**, which is lighter, faster, and answers a what-if while the patient is still in the room
A physics-constrained hybrid Twin features plus an ODE-consistency loss term	**+3.59 mg/dL** against gradient boosting, interval +2.51 to +4.76 — the constraint cost accuracy	The twin as an **independent second engine**, cross-checking every what-if rather than being folded into the forecaster
Three further interval methods Split conformal, conformalised quantiles, adaptive conformal	Coverage falls to **78.1 %** and **75.6 %** in the second cohort; the adaptive method hits its target by returning an **infinite interval** up to 44 % of the time	**CV+**, which holds 80.3 % or better at every horizon in both cohorts and always returns a finite range
Four more ways to personalise From scratch, residual, shrinkage, windowed least squares	Training on one patient alone reaches **18.31 mg/dL** against 16.06 for no personalisation at all, and does not recover inside a week	**Fine-tuning from the population model**, plus continuous least squares for patients with no history yet
Training on the synthetic Indian cohort Adding 100 virtual patients per setting to the training data	**+0.25 to +0.35 mg/dL** — no gain worth having from synthetic patients in training	The synthetic cohort as a **stress test only**, never in training, and labelled SIMULATED wherever it appears
Sleep and heart-rate-variability features Tested at four effect strengths and two study sizes	**At most 0.09 mg/dL** at 9,000 patient-days with the effect inflated five-fold — intervals including zero	The channel **collected and shown, but no claim made on it**, and a sample-size bound published for anyone attempting it

ONE RULE, APPLIED EVERY TIME

The alternative was built before it was rejected

Nothing here was dismissed on reputation or on cost. Each was implemented, trained on the same patients and scored against the same baselines, so the interval beside it is **a measurement, not an opinion**.

WHERE THE WORK WENT

Most of the work is in the six rows above

The served system is one path through a much larger search. **The rejected branches are what make the chosen one defensible.**

HOW EACH ONE WAS DECIDED

Nothing here was tuned into existence

Every row was decided on **inner-validation patients**, against a rule written before the run. The holdout was opened once, at the end, and the numbers it returned are the ones on these slides.

Source: README §9 · docs/DECISIONS.md D-040, D-041, D-051, D-052, D-061, D-064, D-066, D-072 · reports/metrics/hybrid_neural.json, uncertainty.json, personalize_*.json, india_*.json

What the evidence covers, and what comes next

Every limit below is one we measured and recorded rather than discovered late, and each is paired with **the specific piece of work that closes it**. The code for most of them is already written; what is missing is data that does not exist in the open yet.

145	2	1	0	0
real patients across two open cohorts and two countries	independent cohorts — **forecasts, ranges and alerts reported in both**	holdout, **opened once**, at the end	patients **from India** in the real data — the gap that matters most	prospective trials. This is **decision support, not a medical device**

EACH LIMIT, AND THE WORK THAT CLOSES IT

WHAT THE EVIDENCE DOES NOT YET COVER	WHAT CLOSES IT
No Indian patients in the real data. The cohorts are American and Chinese adults; the Indian results are simulated.	A **50 to 100 patient CGM and meal-log study** in an Indian clinic. The three-day local-calibration recipe is already measured and the console already supports it.
ML what-if sizes run low. On matched meals the ML models predict 18.6 to 19.2 of a 35.3 mg/dL rise, the twin 27.5 of 33.5; neither predicts single meals well.	**Richer meal capture** — portion photographs and timestamps rather than recalled grams — then re-running the dose-response check against its pre-registered range.
Sensor bias is not caught by the quality score. A +20 mg/dL bias drops 80 % interval coverage to about 65 %.	A **drift detector** comparing CGM against fingerstick calibrations, and a coverage alarm surfaced on the data-quality page rather than buried in a report.
Coverage is not uniform across subgroups. 73 % on type 2 rows against 87 % on healthy rows, against an 80 % target.	**Group-conditional conformal prediction**, so the 80 % promise holds within each subgroup rather than only on average.
Hypoglycaemia evidence is thin. Only 8 CGMacros patients had any event at all, so that interval is wide.	Pooling hypoglycaemia events across cohorts and adding one with greater hypo exposure **before any hypoglycaemia claim is made**.
Federated and sleep results are simulations. The clinics are groups of open-cohort patients, not real sites.	Running the federated protocol **across two real sites**. The code is written, the privacy accounting is in place, and only the partners are missing.

THE ORDER THIS HAPPENS IN

AVAILABLE NOW

Console and repository, open

Live in a browser with **no install and no sign-in**, every figure traceable to a file.

NEXT

Close the measurement gaps

Group-conditional intervals and the **sensor-drift detector** — both are code, not data.

THEN

Indian pilot data

A small local cohort turns the **simulated India results into measured ones**.

WHAT WE ARE LOOKING FOR

One clinic, and one more

An Indian diabetes clinic willing to run CGM and a meal log for a month, and a **second site to federate with**. That is the whole ask.

BEFORE ANY CLINICAL USE

Prospective validation

A planned study and a regulatory pathway. **Nothing here substitutes for that.**

SCOPE, STATED ONCE

GlucoTwin is **decision support, not a medical device**. It has not been validated prospectively, the demo runs on **120 simulated patients** with a SIMULATED badge on every screen, and the disclaimer appears on every page of the console.

Source: README §15 Limitations and responsible AI · docs/limitations.md (including the DPDP Act mapping) · docs/DECISIONS.md

What this project establishes

Six findings, measured on **held-out real patients from two independent cohorts**, each carrying a confidence interval, and each reproducible from the repository. Together they describe a glucose digital twin that is **ready to be tested in a clinic** rather than only in a notebook.

15.70 mg/dL at +60 min

A forecast that holds in two countries

19 % better than repeating the last reading, and replicated at **15.51** in a second cohort on another continent. On the Clarke Error Grid (development model, CGMacros), **99.6 %** land in zones A or B at +30 min and 98.3 % two hours out.

Beats a per-patient autoregression, a CGM-only model and a fused model — all built, all scored on the same patients.

80.5 % coverage, 80 % promised

Uncertainty that keeps its promise

The band holds at **80.3 % or better** at every horizon in both cohorts, and the same method covers **95.1 to 95.3 %** at the 95 % level — so it is calibrated, not tuned to one number. Band width is printed beside coverage, so a narrow band cannot be mistaken for a good one.

Three of four conformal methods fail somewhere. The one that holds everywhere is the one served.

83 % of episodes caught

Calibrated risk instead of alarms

At the same alert load as the alert a clinic runs today — which catches **59 %**. Risk is still ranked well **two hours ahead** (AUROC 0.77 to 0.82), and a stated chance is within **2.5 points** of the observed rate.

Eighteen dedicated classifiers, two calibrators compared, and the whole threshold curve handed to the clinic.

14.94 mg/dL after seven days

A twin that adapts to one person

Down from **16.06** as it watches a single patient, and still improving at the end of the week. A second method pays from the **very first reading**, so a new patient benefits before any history exists. Six strategies were run over seven days each to choose these two.

Training on that patient alone reaches 18.31 — the population prior is what makes personal data useful.

100 % of carbohydrate-direction checks

A what-if with two engines

The monotone model never lets less carbohydrate predict a higher peak. The physiological twin sizes the change: **0.33 mg/dL per g**, inside the pre-registered range, and **27.5 of 33.5 mg/dL** on matched real breakfasts, on average.

The twin runs live in the browser, corrected by every reading. The walk effect is assumed and labelled.

0.09 mg/dL — the cost of privacy

A route to deployment in India

Federated training across five clinics costs **0.09 mg/dL** against pooling every record centrally, with **no patient record leaving the building**. Local calibration recovers the **3.4 mg/dL** cost of moving a model to a new country.

Tested on a synthetic Indian cohort built from IFCT 2017 meals and published clamp data.

71 of 75
decisions **written before the runs they govern**

33 of 33
result files carry their **commit, data hash and seed**

212
unit tests, **run by one command**

37
configurations **built and scored**; one of them is served

Live
the console runs in a browser — **no install, no sign-in**

Every figure above traces to reports/metrics/*.json and to a numbered decision in docs/DECISIONS.md · cleopatrakilledceaser.github.io/SynapseSquad_IITGuwahati